

Cliff Sun

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EDUCATION

University of Illinois Urbana-Champaign

B.S. Physics, B.S. Mathematics, Minor in Computer Science

Champaign, IL

GPA: 3.94/4.0; Expected May. 2027

RESEARCH POSITIONS

Johns Hopkins University Applied Physics Laboratory

Research Intern, advised by Dr. Michael Wall

Laurel, MD

May 2025 – Present

- Tensor networks, scaling quantum circuits, non-inertial quantum sensing, machine learning

University of Illinois Urbana-Champaign

Undergraduate Researcher, advised by Professor Alexey Bezryadin

Champaign, IL

Aug. 2023 – Present

- Theoretical superconducting quantum hardware with 1D nanowires

AWARDS & HONORS

2026 Barry M. Goldwater Scholarship (400 students selected nationwide)	2026
2026 Astronaut Scholarship (70 students selected nationwide)	2026
Texas Instruments Research Scholar	2025
Illinois Scholars Undergraduate Research Fellow	2025
Best Undergraduate Poster, Chicago Quantum Exchange	2025
Selected presentation at Johns Hopkins APL (4/400 interns selected)	2025 & 2026
Best Technical Presentation, NASA Human Lander Challenge	2025
Finalist, NASA Human Lander Challenge	2024 & 2025
2nd place, NASA Human Lander Challenge	2024

PUBLICATIONS

First-author

5. Sun, C., Forrest, T. K., *et al.*, "Simulating Diffusion-advection via Matrix Product States," *Under preparation* (2026).
4. Sun, C., & Wall, M., "A Machine Learning Architecture for scaling Holographic Quantum Circuits," *Finished; To be submitted to International Conference on Learning Representations (ICLR)* (2026).
3. Sun, C., Bezryadin, A., "A Dayem Loop Qubit Based on Interfering Superconducting Nanowires," *Superconductor Science and Technology* (2026).
2. Sun, C., Zhao, Z., Bezryadin, A., "Perfect Superconducting Diode and Supercurrent Range Controller," *Physics Letters A* (2025).
1. Sun, C., Bezryadin, A., "Multiple-Nanowire Superconducting Quantum Interference Devices: Critical Currents, Symmetries, and Vorticity Stability Regions," *Nano Express* (2025).

Co-author

1. Zhao, Z., Sun, C., Huang, C., Zhang, J., Song, X., & Bezryadin, A., "Symmetry of the critical current function in superconducting nanodevices," *Submitted* (2026). **(Experimental verification of a prediction I made)**

PATENTS

1. Bezryadin, A., Sun, C., "Nanowire-based Superconducting Qubit Device" *Filed U.S. Patent # 64/072,143* (2026).

PRESENTATIONS

Talks

- "Superconducting Nanowires for Digital and Quantum Logic"* Aug. 2026
Astronaut Scholarship Technical Conference; Primary Presenter
- "A Machine Learning Architecture for scaling Holographic Quantum Circuits"* Aug. 2026
Johns Hopkins Applied Physics Laboratory; Primary Presenter
- "Simulating Diffusion-Advection via Matrix Product States"* Aug. 2026
Johns Hopkins Applied Physics Laboratory; Primary Presenter
- "Quantum algorithms, circuits, and sensors"* Aug. 2026
Johns Hopkins Applied Physics Laboratory; Primary Presenter
- "Superconducting Nanowires for Digital and Quantum Logic"* Apr. 2026
UIUC Undergraduate Research Symposium; Primary Presenter
- "Holographic Quantum Tensor Networks and Quantum Annealing"* Aug. 2025
Johns Hopkins Applied Physics Laboratory; Primary Presenter
- "ECLIPSE: Efficient Cryogenic Low Invasive Propellant Supply Exchange"* June 2025
NASA Marshall Space Flight Center; Primary Presenter
- "HINDER: Holistic Integration of Navigational Dynamics for Erosion Reduction"* June 2024
NASA Marshall Space Flight Center; Co-Presenter

Posters

- "Superconducting Quantum Interference in Nanowire Networks for Digital and Quantum Logic"* Mar. 2026
APS Global Summit; Primary Presenter
- "Multiple-Nanowire Superconducting Quantum Interference Devices: Critical Currents, Symmetries, and Vorticity Stability Regions"* Nov. 2025
Chicago Quantum Exchange; Primary Presenter
- "Model-fitting Algorithm for Disordered Josephson Junction Arrays"* Sept. 2024
UIUC Engineering Exposition; Primary Presenter
- "Analyzing Disorder and Symmetries in Random Josephson Junction Arrays"* Apr. 2024
UIUC Undergraduate Research Symposium; Primary Presenter

TEACHING, OUTREACH, & LEADERSHIP

- UIUC Office of Undergraduate Research, Undergraduate Research Ambassador** 2024 – Present
- Leading workshops, symposiums, and panels
- NASA Human Lander Challenge, NASA Competition Lead** 2024-2025

- Led design of systems-level cryogenic propellant transfer architecture; awarded \$9k; now used by NASA

MEDIA FEATURES

Illinois Physics News	June 2026
<i>"Illinois Physics undergraduate student Cliff Sun named a 2026 Astronaut Scholar"</i>	
Illinois News Bureau	April 2026
<i>"Four Illinois students receive Goldwater scholarships"</i>	
The Daily Illini	January 2026
<i>"UI students, alumni reflect on published works"</i>	
Chicago Quantum Exchange	November 2025
<i>"Five researchers recognized with poster awards at eighth annual Chicago Quantum Summit"</i>	
Illinois MechSE	April 2025
<i>"Interdisciplinary student team named finalist in NASA Human Lander Challenge"</i>	
Illinois Aerospace Engineering	July 2024
<i>"Undergrad team wins 2nd place in national NASA competition"</i>	